

Coding Challenge 03

Project Plan & 5-Member Task Distribution

INTEGRATED CONTROL STATEMENTS & DECISION LOGIC

Three Python programs · Testing evidence · One final ZIP submission

Project Overview

AREA	ASSIGNMENT OUTPUT	OWNER
Challenge 1	Loan Decision Engine	Member 1
Challenge 2	Quadratic Equation & Parabola Analyzer	Member 2
Challenge 3	Two-Variable Linear System Analyzer	Member 3
Testing + Evidence	All three challenges + required PDF testing report	Member 4
Integration + Final QA	Integration, documentation, filenames, ZIP and final checks	Member 5

What Must Be Submitted

- **CC03_Challenge01_LoanDecisionEngine.py** — complete working Challenge 1 program.
- **CC03_Challenge02_QuadraticAnalyzer.py** — complete working Challenge 2 program.
- **CC03_Challenge03_LinearSystemAnalyzer.py** — complete working Challenge 3 program.
- **CC03_Testing_Report_LastName_FirstName.pdf** — testing evidence for all three challenges.
- **Final ZIP:** APPDAET_CodingChallenge03_LastName_FirstName.zip. The ZIP should contain exactly the four required files and no unrelated files.

01 | Member 1 — Loan Decision Engine

- **Inputs:** Monthly Income, Monthly Debt, Credit Score, Employment Duration, Requested Loan, Existing Customer (yes/no).
- **Validation:** income > 0 ; debt ≥ 0 ; credit score 300–850; employment duration ≥ 0 ; requested loan > 0 ; customer must be yes/no.
- **Calculations:** $DTI = \text{debt} / \text{income} \times 100$; maximum recommended loan = income $\times 12$.
- **Credit classification:** Excellent 750–850; Good 700–749; Fair 650–699; High Risk 300–649.
- **DTI classification:** Low $\leq 30\%$; Moderate $> 30\%$ to 40% ; High $> 40\%$ to 50% ; Very High $> 50\%$.
- **Decision rules:** High Risk $\rightarrow$ Declined; $DTI > 50\% \rightarrow$ Declined; employment < 6 months $\rightarrow$ Declined; qualifying Excellent/Good cases $\rightarrow$ Approved or For Review; qualifying Fair cases $\rightarrow$ For Review; otherwise Declined.
- **Final report:** show inputs, DTI and classification, credit classification, maximum recommended loan, requested amount status, final decision, and reason.

02 | Member 2 — Quadratic Analyzer

- Accept a, b, c for $ax^2 + bx + c = 0$ and validate that $a \neq 0$.
- Calculate $D = b^2 - 4ac$ and classify: $D > 0$, $D = 0$, or $D < 0$.
- Calculate real roots when applicable; identify repeated root when $D = 0$.
- Calculate axis of symmetry $x = -b/(2a)$, vertex, opening direction, and y-intercept (0,c).
- Report whether the vertex is a minimum or maximum based on the sign of a.
- Use the required output labels so results are understandable without reading the code.

03 | Member 3 — Linear System Analyzer

- Accept coefficients for $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$.
- Reject an equation when both variable coefficients are zero.
- Calculate D, D_x , and D_y using the required determinant formulas.
- Classify the system as unique solution, no solution, or infinitely many solutions.
- For a unique solution, calculate x and y and verify them by substitution into both original equations.
- Test negative and decimal coefficients as required.

04 | Member 4 — Testing & Evidence

Member 4 owns the testing matrix and the single PDF testing report required by the submission instructions. Testing must cover multiple cases, not just one successful run.

- **Challenge 1:** approved, for review, declined, invalid input, boundary value, and more than one decline reason where applicable.
- **Challenge 2:** positive discriminant, zero discriminant, negative discriminant, upward-opening parabola, downward-opening parabola, and invalid quadratic input.
- **Challenge 3:** unique solution, no solution, infinitely many solutions, negative coefficients, decimal coefficients, and invalid equation input.
- **Evidence for each case:** challenge title, test case number, input values, expected result, actual output, pass/fail statement, and screenshot or clearly captured console output.
- **Required report filename:** CC03_Testing_Report_LastName_FirstName.pdf.

05 | Member 5 — Integration, Documentation & Final QA

- Collect the three completed Python programs and the testing report.
- Check source-code headers: Course, Coding Challenge 03, challenge title, student name, student number, section, date, and program purpose.
- Review validation, formulas, Boolean conditions, decision-rule order, boundary handling, readability, and output labels.
- Confirm the required filenames exactly match the submission instructions.
- Confirm the ZIP contains exactly four required files and no unrelated files.
- Open the ZIP before submission to make sure it is valid.
- Perform a final checklist with all five members before uploading.

Team Workflow

STEP	OWNER	HANDOFF
1 · Build	Members 1–3	Working Python programs
2 · Test	Member 4	Expected vs. actual results + screenshots
3 · Fix	Members 1–3	Corrections from QA
4 · Integrate	Member 5	Final files + report
5 · Verify	All 5 members	Final ZIP and submission check

Final Submission Checklist

- All three challenges are complete and working.
- Input validation is implemented before dependent calculations.
- Mathematical formulas and signs have been checked.
- Decision logic and boundary rules have been checked.
- Required test cases are completed.
- Testing report includes expected results, actual output, pass/fail, and screenshots.
- Source-code headers are present.

- Output is clearly labeled and understandable.
- Filenames are exact.
- ZIP contains exactly the four required files.
- ZIP opens correctly before uploading.

Core principle: Accept data → validate data → calculate → evaluate conditions → select the correct decision path → produce an understandable result → verify through testing.